

# Bubbles in a ferromagnetic superfluid

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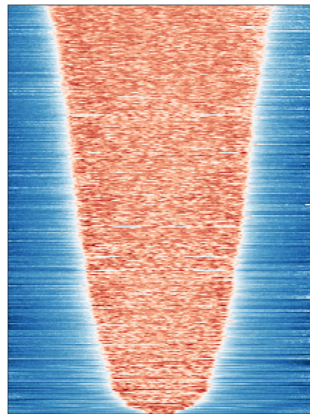


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This presentation will cover:

- ▶ **Introduction**
- ▶ **Theoretical background:** Ferromagnetism in coherently coupled two-component spin mixtures
- ▶ **Data analysis:** Characterization of false vacuum decay bubbles
- ▶ **Conclusions**



Why bubbles in a ferromagnetic superfluid?

- ▶ First **experimental observation** of false vacuum decay (FVD) via bubble formation in the *Pitaevskii BEC Center* laboratories of the University of Trento.
- ▶ **Superfluidity**: high degree of coherence of the system
- ▶ **Ferromagnetism**: double well energy landscape
- ▶ Study of FVD provides information on metastability, from quantum systems to cosmology

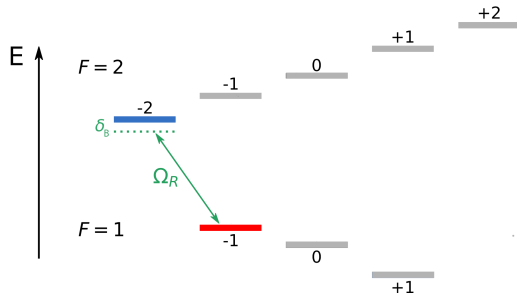
# Experimental platform

- $N \sim 10^6$  condensed  $^{23}\text{Na}$  atoms in the hyperfine states:

$$|F = 2, m_F = -2\rangle = |\uparrow\rangle$$

$$|F = 1, m_F = -1\rangle = |\downarrow\rangle$$

- Contact interaction constants  $g_{\uparrow\uparrow}$ ,  $g_{\downarrow\downarrow}$ ,  $g_{\uparrow\downarrow}$
- Interconversion due to Rabi coupling: strength  $\Omega_R$  and detuning  $\delta_B$
- Harmonic trapping potential (cigar-shaped, 1D)





# Coherently coupled two-component spin mixtures

The GPEs contain the **intra** and **inter**-species interaction constants and the **coupling** between the states:

$$\begin{aligned} \left[ -\frac{\hbar^2}{2m} \nabla^2 + V(x) - \frac{\delta_B}{2} + g_{\uparrow\uparrow} |\psi_{\uparrow}(x)|^2 + g_{\uparrow\downarrow} |\psi_{\downarrow}(x)|^2 \right] \psi_{\uparrow}(x) - \frac{\hbar\Omega_R}{2} \psi_{\downarrow}(x) &= \mu_{\uparrow} \psi_{\uparrow}(x) \\ \left[ -\frac{\hbar^2}{2m} \nabla^2 + V(x) + \frac{\delta_B}{2} + g_{\downarrow\downarrow} |\psi_{\downarrow}(x)|^2 + g_{\downarrow\uparrow} |\psi_{\uparrow}(x)|^2 \right] \psi_{\downarrow}(x) - \frac{\hbar\Omega_R}{2} \psi_{\uparrow}(x) &= \mu_{\downarrow} \psi_{\downarrow}(x) \end{aligned}$$

# Coherently coupled two-component spin mixtures

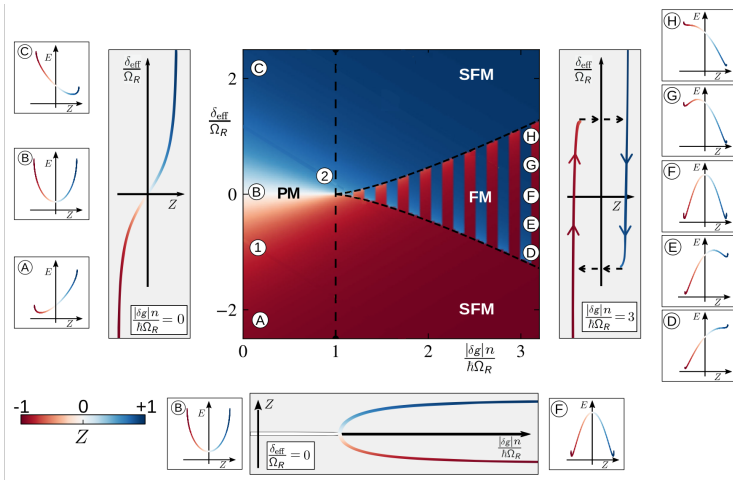
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Magnetization  $Z = (n_{\uparrow} - n_{\downarrow}) / (n_{\uparrow} + n_{\downarrow})$  produces a double well energy landscape:

$$E_{\text{MF}}(Z) = -|\delta g| n Z^2 - 2\hbar\Omega_R \sqrt{1 - Z^2} - 2\hbar\delta_{\text{eff}} Z$$

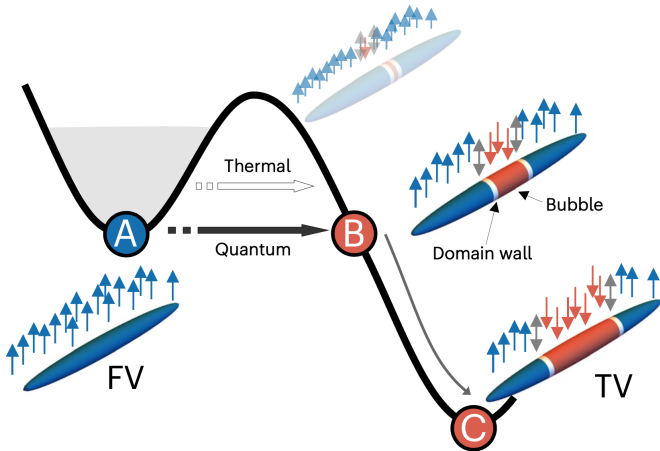
# Magnetic model phase diagram: ground state configuration



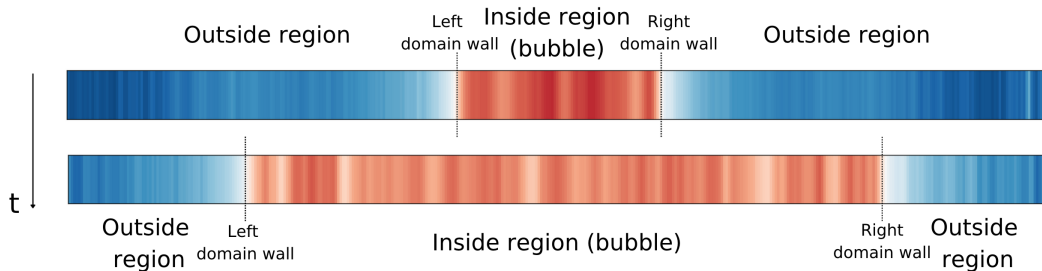
- Set  $\frac{|\delta g|n}{\hbar\Omega_R}$  to a fixed value by adjusting  $\Omega_R$
- Probe the magnetic properties by changing  $\delta_{\text{eff}}$

# False vacuum decay bubbles

- Quantum tunnelling from A to B (stochastic)
- Decay from B to C (what we want to study)
- Problem: when to take the shot?



# Example of bubble shots

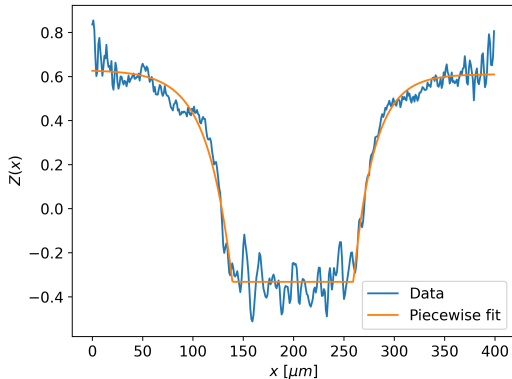
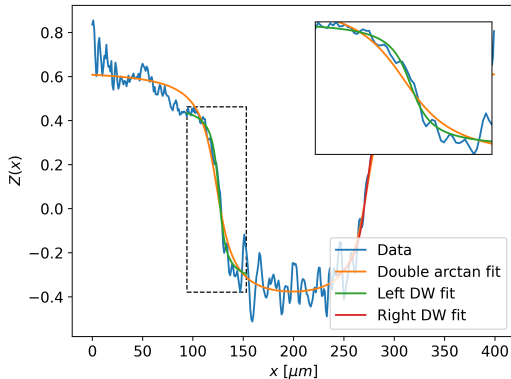


Our aim is to study:

- ▶ Size  $\sigma_B$  and domain wall width  $w_D$
- ▶ Excitation structures inside/outside

# Bubble fitting routines

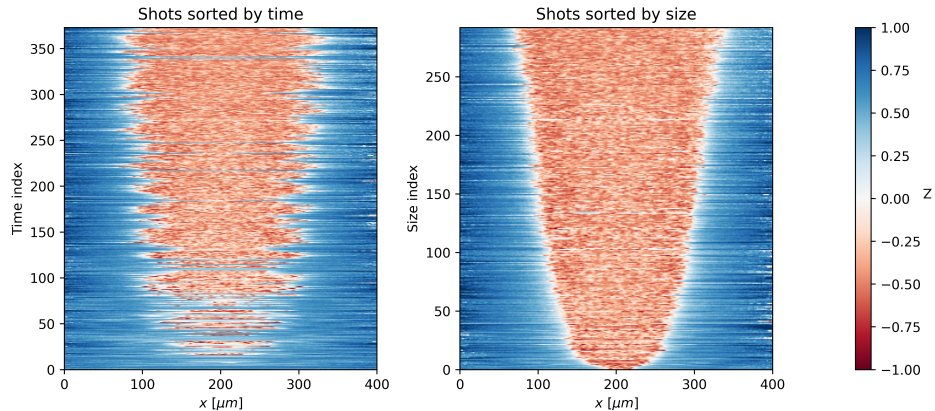
Example of fitting routines,  $\Omega_R/2\pi = 400$  Hz and  $\delta = 596.5$  Hz



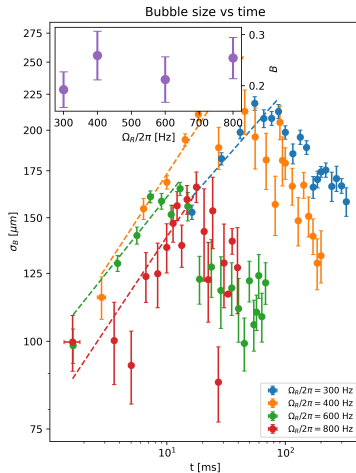
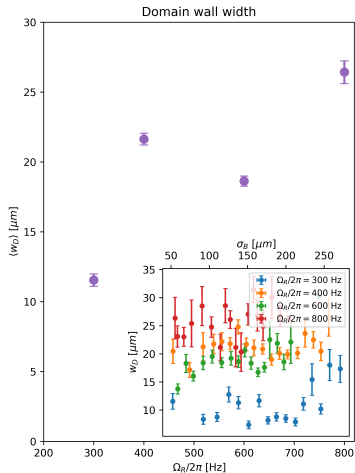
# Shot sorting



Bubble shots with  $\Omega_R/2\pi = 400$  Hz and  $\delta = 596.5$  Hz



# Analysis of domain wall width and bubble size



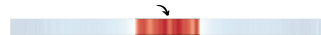
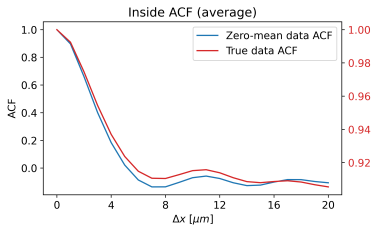
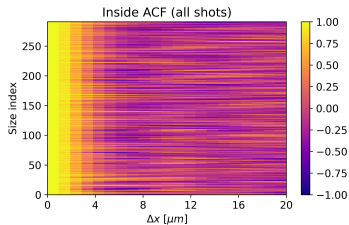
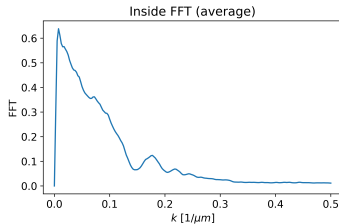
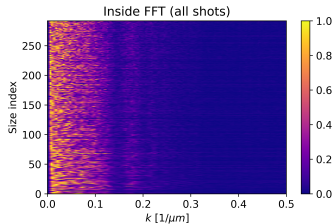
- Shots clustered by size (left) or time (right)
- Fit function:

$$\sigma_B(t) = A \left( \frac{t}{1 \text{ ms}} \right)^B$$



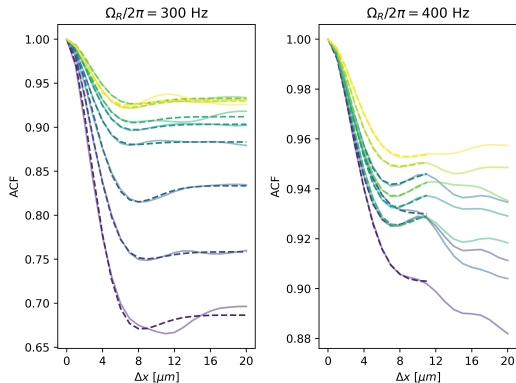
# Spectral analysis in the inside region

FFT and ACF on shots with  $\Omega_R/2\pi = 400$  Hz and  $\delta = 596.5$  Hz



- FFT: broad peak at  $k \sim 0.01 \mu\text{m}^{-1}$
- ACF: peak at  $\Delta x \sim 11 \mu\text{m}$

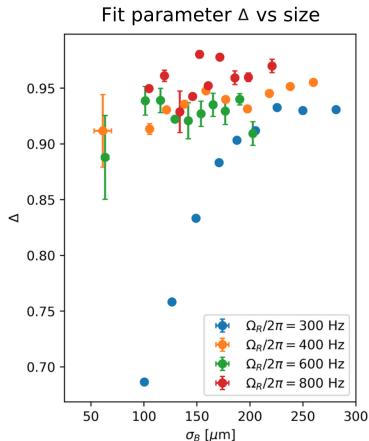
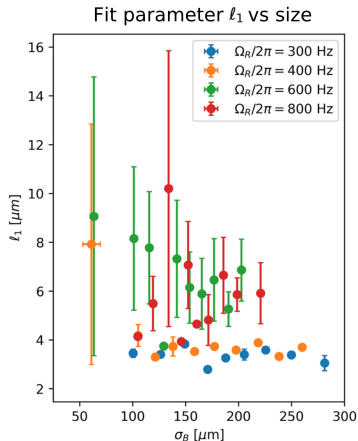
# ACF analysis in the inside region vs $\sigma_B$



- ▶ Shots clustered by increasing size (darker to lighter colors)
- ▶ Fit function  $\mathcal{A}_{\text{fit}}(x) =$

$$(1-\Delta) \cos\left(\frac{\pi x}{\ell_2}\right) \exp\left[-\frac{1}{2} \left(\frac{x}{\ell_1}\right)^{1.7}\right] + \Delta$$

# ACF analysis in the inside region vs $\sigma_B$



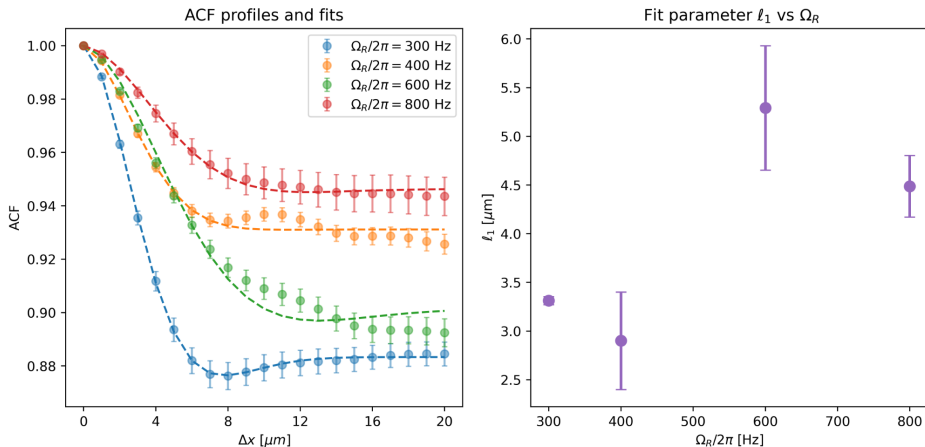
Healing lengths:

$$\xi_s = \frac{\hbar}{\sqrt{2mn|\delta g|}} \approx 0.5 \mu\text{m}$$

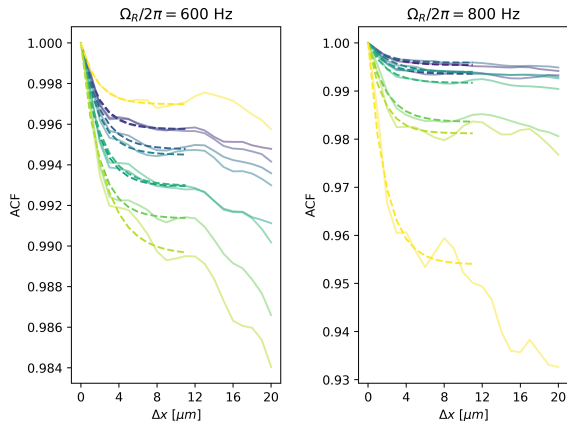
$$\xi_d = \frac{\hbar}{\sqrt{2mn\bar{g}}} \approx 0.01 \mu\text{m}$$

$$\xi_R = \sqrt{\frac{\hbar}{m\Omega_R}} \approx 1.8 - 3.0 \mu\text{m}$$

# ACF analysis in the inside region vs $\Omega_R$



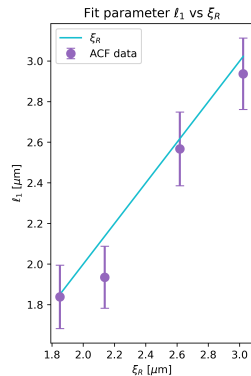
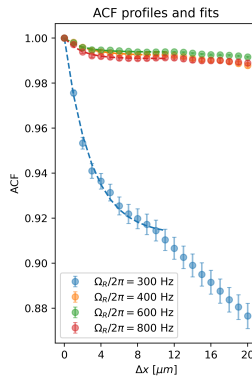
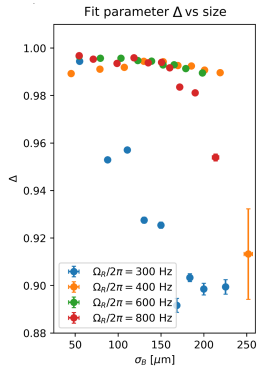
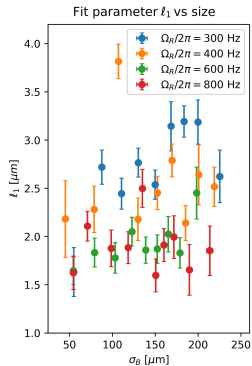
# ACF analysis in the outside region vs $\sigma_B$



- ▶ Shots clustered by increasing size (darker to lighter colors)
- ▶ Fit function:

$$\mathcal{A}_{\text{fit}}(x) = (1 - \Delta) \exp\left[-\frac{x}{2\ell_1}\right] + \Delta$$

# ACF analysis in the outside region vs $\sigma_B$ and $\Omega_R$



# Conclusions

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What did we learn?

- ▶ System shows **different properties** between inside and outside of the bubble
- ▶ Domain wall width **depends on the coupling** strength  $\Omega_R$
- ▶ Growth factor of the bubble size in time is **independent** of  $\Omega_R$
- ▶ In the bubble, periodic structures **disappear** with size increasing. They **appear**, instead, outside of the bubble.
- ▶ Length scale of information outside is related to the Rabi **healing length**

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Future research:

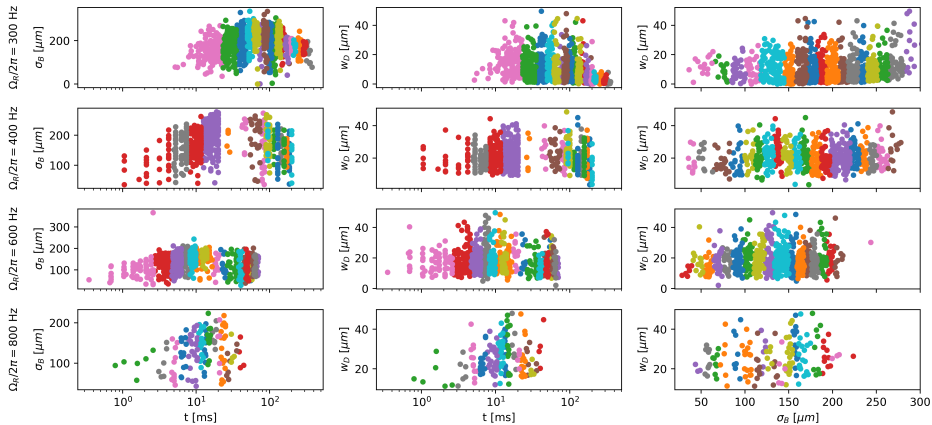
- ▶ Analysis of the **density** channel
- ▶ Comparison with numerical GPE **simulations**
- ▶ Behavior at different **temperatures**

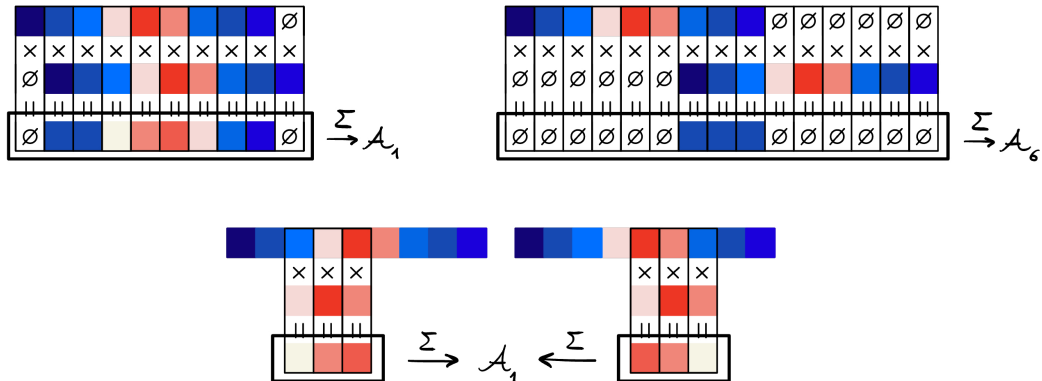


Thank you for your attention!

# Data clustering

## Clustering





# FFT and ACF averages in the inside region

